

Sina Barazandeh

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Summary

Ph.D. student in Carnegie Mellon's School of Computer Science, building ML systems and AI agents that run experiments. Led CMU's Model Hardware Standard deployment—instrument drivers and orchestration for closed-loop experiments—featured in [Anthropic's research preview](#) and [Nature News](#). Also built generative sequence models, including first-author UTRGAN. Seeking Summer 2027 internships in ML systems, reinforcement learning, sim-to-real, and generative models.

Education

Carnegie Mellon University

Pittsburgh, PA

Ph.D. Student, School of Computer Science | Advisor: Jose Lugo-Martinez

Aug 2024 – Expected May 2029

- Program: Computational Biology. Coursework: Machine Learning (10-701), LLM Systems (11-868)

Bilkent University

Ankara, Turkey

M.Sc., Computer Engineering | GPA: 3.9/4.0 | Advisor: A. Ercument Cicek

Sep 2021 – Jun 2024

- Thesis: Generative Models for Generating and Optimizing Biological Sequences

Shiraz University

Shiraz

B.Sc., Computer Engineering | GPA: 3.5/4.0

Sep 2016 – Jun 2021

Coursework: Parallel Programming, Computer Graphics (OpenGL)

Research Experience

Carnegie Mellon University, School of Computer Science

Pittsburgh, PA

Graduate Research Assistant | Advisor: Jose Lugo-Martinez

Aug 2024 – Present

- Led CMU's MHS deployment—featured in [Anthropic's research-preview announcement](#) and [Nature News](#)—building drivers and orchestration across a liquid handler, plate reader, robotic arm, and cameras; experiments ran about 3× faster than the prior workflow
- Developing **simPCL**, a simulator and benchmark generator for agent planning and fault recovery in programmable cloud laboratories; presented related work at Microsoft Research's AI Scientist Summer Workshop
- Developing proximal policy optimization for senescence biomarker discovery

Bilkent University, Department of Computer Engineering

Ankara, Turkey

Research Assistant, CicekLab | Advisor: A. Ercument Cicek

Sep 2021 – Jun 2024

- Developed GANs and sequence-to-sequence transformers for RNA design: first-author **UTRGAN** (5' UTR optimization); co-author on **RNAttranslator** and **RNAGEN**

Technical Skills

ML/AI: PyTorch, JAX; reinforcement learning, sim-to-real, generative models, LLMs; CUDA, multi-GPU and distributed training

Simulation: MuJoCo; 3D simulation and world models

Computational biology: single-cell RNA-seq; protein language models

Systems: Python, C/C++; Git, Docker, Singularity; SLURM, HPC clusters

Selected Publications

- S. Barazandeh, F. Ozden, A. Hincer, U.O.S. Seker, A.E. Cicek. "UTRGAN: Learning to generate 5' UTR sequences for optimized translation efficiency and gene expression." *Bioinformatics Advances*, 5(1): vbaf134, 2025.
- F. Ozden, S. Barazandeh, et al. "RNAGEN: A generative adversarial network-based model to generate synthetic RNA sequences to target proteins." *ChemBioChem*, 27(13): e202500445, 2026.
- S.S. Tabrizi, S. Barazandeh, H.H. Aghdam, A.E. Cicek. "RNAttranslator: Modeling protein-conditional RNA design as sequence-to-sequence natural language translation." *PLOS Computational Biology*, 21(10): e1013541, 2025.

Awards & Achievements

Second Best Presentation Award, HIBIT Conference, Ankara, Turkey

2023

Comprehensive Graduate Scholarship, Bilkent University (Full tuition + stipend)

2021–2024

8th Place (32 teams), RoboCup Soccer 2D Simulation League, Montreal, Canada

2018

Full National Scholarship, Shiraz University

2016–2021

Additional Experience

Teaching Assistant | CMU & Bilkent University

2021 – 2026

Courses: Machine Learning, Artificial Intelligence, Programming and Algorithms, ML for Scientists, Computational Medicine

Co-Founder & Backend Developer | Ghasedak Institution, Shiraz

2019 – 2022

Developed the backend for an educational e-commerce platform

Peer Reviewer (2022–2026) | *Bioinformatics* and *Bioinformatics Advances*